

**Brave Today**

The new privacy-friendly news reader app prevents parties from tracking sites, articles and topics people view. <http://dgit.in/jan21-69>

**Farting Through The Streets**

Tesla drivers can now customize their horn to fart noises, and other meme-ish sounds, thanks to new update. <http://dgit.in/jan21-70>

Facial recognition tech in the new normal



Face masks and facial recognition technology are both pervasive now, but how will they coexist?

Dhriti Datta | dhriti@digit.in

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s people across the globe don masks in order to protect themselves and others from the

onslaught of COVID-19, the promising facial recognition technology that was being implemented in various places, from airports to most modern-day smartphones, has taken a hit. A few countries have even made wearing masks in public mandatory in order to curb the incessant spread of the virus that has been plaguing the globe. Naturally, facial recognition algorithms are encountering

problems with identifying faces with people wearing masks since they cover a large portion of the face, including some vital features such as the nose, mouth and chin. Most facial recognition algorithms are designed to largely factor these parts of the face, and even some of the most robust facial recognition systems are showing signs of failure due to the challenges posed by facial masks. Try unlocking a smartphone with a face mask on and you will, most likely, encounter this issue.

Recently, a preliminary study from the National Institute of Standards and Technology (NIST) analysed how facial recognition technologies were faring with people wearing masks.



Examples of photos that were digitally masked in the NIST study

According to the study, the masks did prove to be a major hurdle for some facial recognition systems. The study analysed 89 commercial facial recognition algorithms, including ones from Samsung and Panasonic, and studied their performance on images of approximately one million individuals. A group of photos were then “digitally masked” by adding superimposed shapes in various colours on faces to mimic masks. These shapes covered the subject’s nose, mouth and chin, as masks would. The study found that wearing face masks reduces the overall efficacy of facial recognition systems. According to NIST, the 89 commercial facial recognition algorithms tested had error rates between 5 and 50 per cent when matching the digitally masked faces to the photos of the same individuals without masks. While this would be the expected outcome under normal circumstances, solutions are now being sought for facial recognition systems to be able to securely identify a face even when the face is partially obscured.

RIISING UP TO THE CHALLENGE

Some companies and researchers have already begun making strides in overcoming this situation by building new datasets of masked images for the algorithms to learn and upgrading their existing solutions. For example, NEC Corporation, a Japanese multinational information technology and electronics company, announced the development of a touchless, multimodal biometric authentication system which utilises facial features and iris recognition technologies in conjunction to make the system more reliable. Dubbed the Bio-IDiom system, the system is alleged to reli-